

SEBASTIAN JARAMILLO DIAZ

DATA SCIENCE | FINANCIAL ANALYTICS | PYTHON, SQL

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PROFESSIONAL SUMMARY

Quantitative researcher with doctoral training in Mathematics and an MSc in Economics, combining deep statistical expertise with applied experience in financial analytics. Develops Python-based analytical solutions for credit risk, financial forecasting and portfolio analysis, with careful attention to the business question and the reliability of the underlying data. Experienced in taking ownership of complex problems from formulation through implementation and validation. Brings sound judgment about model assumptions and uncertainty, alongside the ability to explain findings clearly and work collaboratively. Committed to building practical, well-supported solutions that inform financial decisions.

CORE SKILLS

Programming: Python (pandas, NumPy, SciPy), SQL, R; modular and object-oriented programming.

Machine Learning: scikit-learn, CatBoost, Random Forest, PyTorch, TensorFlow; feature engineering, regularization, model diagnostics and evaluation.

Financial Analytics: Credit risk, delinquency forecasting, bank income forecasting, stress testing, portfolio optimization and hedging.

Statistics and Econometrics: Time series, panel regression, Bayesian inference, hypothesis testing, Monte Carlo simulation and sensitivity analysis.

Data Workflows: Data collection and integration, ETL, quality controls, time-aware validation, leakage prevention and reproducible experiments.

Tools: Git/GitHub, Jupyter, Docker, Excel; technical documentation.

SELECTED DATA SCIENCE AND FINANCE PROJECTS

Credit Risk and Predictive Analytics – Erdős Institute

Spring 2026

- Built an end-to-end Python pipeline integrating NOAA severe-weather records with county-level mortgage data; collected, cleaned and joined 73,899 county-month observations and engineered 60 predictive features.
- Trained CatBoost to forecast next-month early delinquency, reducing held-out RMSE by 21% against a persistence baseline. Applied data-quality controls, temporal validation, feature-importance analysis and two-way fixed-effects regression.

Banking Forecasts and Stress Testing – University of Minnesota

Summer 2024

- Collaborated with a data scientist at the Federal Reserve Bank of Minneapolis to forecast bank interest income for stress testing, integrating 20+ years of FR Y-9C filings with macroeconomic indicators.
- Benchmarked regression, autoregressive and tree-based models; Random Forest achieved out-of-sample $R^2 = 0.87$. Engineered quarterly lag features and evaluated robustness during the 2008 and COVID-19 crises.

Portfolio Optimization and Risk Analysis – Erdős Institute

Summer 2025

- Implemented constrained optimization in Python/SciPy under beta, drawdown and volatility limits. Historical evaluations improved Sharpe ratios from 0.87 to 1.05 and from 0.31 to 0.47 for two portfolio risk profiles.
- Evaluated rolling-window return behavior and delta/vega hedging under Black–Scholes, Heston and GARCH models, comparing risk-adjusted performance and tail exposure.

Scientific Machine Learning for Option Pricing – IMPA

2026

- Developing HestonGuard, a physics-informed operator-learning project for the Heston option-pricing PDE, with classical numerical benchmarks, adaptive sampling and stress scenarios.
- Designed validation around Gamma, a financial sensitivity measure, using prediction-error diagnostics and model-stability checks to identify unreliable outputs.

PROFESSIONAL AND RESEARCH EXPERIENCE

University of Oregon

Doctoral Researcher – Probability and Statistical Modeling

Eugene, OR

Sep. 2021 – Present

- Develop computational algorithms for Markov chains, Monte Carlo methods and MCMC/HMC in Python and R; use simulation experiments to evaluate convergence, dependence and sampling efficiency.
- Build version-controlled research workflows spanning data generation, transformation, modeling, visualization and documentation, using modular notebooks, Git and diagnostic checks.
- Formulate analytical problems, implement numerical experiments and interpret results; communicate mathematical assumptions and computational findings through technical writing, presentations and university instruction.

Handshake

Mathematical Data Analyst

Remote

Jun. 2025 – Sep. 2025

- Validated advanced mathematical solutions against technical evaluation criteria for machine-learning training data, identifying errors, inconsistencies and edge cases.
- Documented recurring failure modes and evaluation guidelines to improve data reliability and consistency in large-language-model training workflows.

Universidad Nacional de Colombia

Quantitative Research and Teaching Assistant

Medellín, Colombia

2016 – 2021

- Supported mathematical research and computational analysis while teaching and explaining complex concepts to audiences with different technical backgrounds.

EDUCATION

University of Oregon

PhD in Mathematics – Probability and Stochastic Processes.

In progress

University of Oregon

MSc in Economics – Econometrics.

Universidad Nacional de Colombia

MSc in Mathematics; BSc in Mathematics.

ADDITIONAL TRAINING AND CREDENTIALS

Erdős Institute: Data Science and Deep Learning Bootcamps; Quantitative Methods in Finance.

IMA / University of Minnesota: Math-to-Industry Bootcamp, 2024.

Society of Actuaries: Exam P – Probability (Passed); Exam FM – Financial Mathematics (Passed); Exam SRM – Statistics for Risk Modeling (Passed).

LANGUAGES

Spanish: Native. | English: Professional working proficiency.